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RESEARCH ARTICLE

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Channel transparency and omnichannel retailing: The impact of sharing retail store product availability information

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Abstract

As more firms look to implement omnichannel strategies to mitigate the inherent information barriers between online and offline channels, understanding the outcomes of these initiatives is critical. This study empirically examines the impact of the decision to share brick-and-mortar (B&M) store product availability information with customers, specifically focusing on the scenario where a discrepancy exists in the product assortment of online and offline channels. We investigate the impact of this information sharing strategy on customer research shopping trajectories using a proprietary dataset collected from a leading North American retailer. Using a difference-in-differences approach, we find that this strategy helps boost the overall sales from customers who reside within the trade area of a retail store. At the channel level, the policy induces customers to migrate from the B&M channel to the online channel after learning that the product selection offered at their local retail store may not meet their needs, resulting in a significant increase in online sales, but a slight decrease in B&M sales. From an operational perspective, our empirical evidence suggests that sharing retail store product availability information leads to more customers using the expedited shipping service when shopping online. Furthermore, we also discover a slight increase in the overall return rate. We conclude by providing managerial implications on the functionality of the online and B&M channels and how retailers can coordinate services offered by both channels to improve the utility customers receive from the brand.

KEYWORDS

information sharing, omnichannel retailing, quasi-experimental methods, retail operations

Highlights

- This study examines the impact of sharing brick-and-mortar (B&M) store product availability on sales, order fulfillment, and product returns in an omnichannel context.

- Using a quasi-field experimental design based on proprietary data collected from a leading North American retailer, this study shows that customers utilize this new information to adjust their shopping plans and avoid unsuccessful shopping trips, leading to increased sales for the brand.
- This study underscores the benefits of improving information transparency between a firm and its customers. It also contributes to the discussion on omnichannel product assortment planning and the future role of B&M stores.

1 | INTRODUCTION

The omnichannel retail environment provides customers with new options to tailor their shopping journey. Based on the information and fulfillment options offered, customers form various shopping trajectories such as "buy online, pickup in-store," "buy in-store, ship to home," webrooming, and showrooming in an attempt to maximize the utility obtained from the shopping experience (Bell et al., 2014). As a result, companies are racing to launch strategies that bridge and integrate their tangible and intangible resources across channels so that shoppers can customize shopping plans to best meet their needs.

Sharing brick-and-mortar (B&M) store product selection information is one of the strategies that has gained popularity among a wide range of retailers, such as Home Depot, Target, and IKEA. This strategy allows customers to check real-time in-store product selection while browsing the retailer's online website. With this additional information, companies hope to provide customers with a frictionless shopping experience across channels and avoid customer dissatisfaction due to unexpected in-store product unavailability. Despite the growing popularity of this information sharing strategy, its impact in the omnichannel context remains vastly understudied, especially from an empirical perspective. Prior studies in this field primarily focus on the effect of disclosing product availability/inventory information in a single channel setting. Furthermore, even with a single-channel retailer, studies have demonstrated diverse impacts of such information depending on the set-up of the retailer, its products, and other contextual factors (Allon & Bassamboo, 2011; Calvo et al., 2020; Cui et al., 2019; Park, Rabinovich, Tang, & Yin, 2020). Thus, it is necessary to explicitly evaluate the impact of this information sharing strategy in the omnichannel context.

Our study contributes to this gap in the literature by exploring changes to customer demand, order fulfillment, and product returns following the rollout of an omnichannel information sharing strategy that allows customers to check real-time in-store product selection while browsing online. An omnichannel retailer may offer different product assortments between its online and B&M channels as well as across different B&M locations, so this new information discloses not only whether a product is available for purchase, but also from which channel(s) and location(s). This is an important question given the rising trend in the retail industry of reducing the size of B&M stores to control operational costs. For example, Macy's and Target both announced plans to transform some of their B&M stores into "neighborhood stores" that are much smaller in size (Kapner, 2018; Kumar, 2020). With B&M stores getting significantly smaller in size, they will inevitably carry fewer stock-keeping units (SKUs), potentially leading to an increase in the gap between the product assortment offered through a retailer's online channel and its B&M stores, making the ability to acquire product selection information upfront more valuable.

To evaluate the impact of this strategy, we collected a proprietary dataset from a nationwide retailer that has an established online and B&M presence. This retailer offers a broader product assortment in its online channel compared to its B&M channel. The exact product selection information at each store was previously unknown to customers unless they visited a retail store. The retailer added a B&M store product availability look-up function to its online website in October 2015. Using this function, customers can check each product's real-time in-store availability information. We obtained point-of-sale data, daily SKU-level inventory records, online order fulfillment data, and product return records for a 39-month period, which covered the retailer's operations before and after implementing the strategy. This data allows detailed measurements of customer purchasing decisions across channels and product selection information at each retail store.

We adopt a quasi-field experimental design to evaluate the demand and operational impact of this information sharing strategy. Using a difference-in-differences (DiD) approach, we find that providing cross-channel product assortment transparency reduces lost sales from customers encountering unexpected product unavailability, because they can adjust their shopping plans upfront to avoid unsuccessful trips. This leads to a 3.5% increase in overall sales. At the channel level, the strategy induces customers to migrate from the B&M channel to the online channel after learning that the product selection offered at their local retail store may not meet their needs, resulting in a 13.1% increase in online sales, but a 2.6% decrease in B&M sales. However, since the online channel is inherently associated with slower order fulfillment and less accurate product fit information, customers more frequently use expedited shipping and exhibit a slightly increased product return rate following the rollout of this strategy.

To the best of our knowledge, this is the first empirical study that addresses the value of providing product availability information in the omnichannel context. A related study by Gallino and Moreno (2014) infers the impact of disclosing B&M store product availability by examining the sales outcomes following the implementation of the "Buy Online, Pick-up in Store" (BOPS) strategy. The logic is that if an item is available for in-store pickup, it implies product availability. However, the BOPS strategy also alters the fulfillment options customers have access to, as customers can reserve their online purchases for later in-store pickup. This makes it more challenging to measure whether customers are attracted to or turned away from the B&M channel based solely on the product availability information. To that end, our study offers a clean research setting that allows us to specifically focus on the value of the product availability information. Furthermore, our study contributes to the retail operations literature by addressing the impact of an omnichannel strategy in the presence of product assortment differentiation between the online and B&M channels, which answers the call by Mou et al. (2018) and Caro et al. (2020) for more research on the operations of smaller retail stores with "tailored assortments."

The remainder of this paper is organized as follows: Section 2 summarizes relevant prior research. Section 3 introduces the institutional setting and develops research hypotheses. Section 4 describes the datasets and econometric design. The main results are reported in Section 5, followed by robustness checks in Section 6 and conclusions in Section 7.

2 | LITERATURE REVIEW

Our study resides within the broad set of literature that explores customer shopping journeys in the omnichannel context. Previous work in this space examines customer strategic reactions to various omnichannel strategies, including initiatives that synchronize information and services across channels and offer new demand fulfillment and return options (Akturk et al., 2018; Akturk & Ketzenberg, 2022; Bell et al., 2014; Bell et al., 2017; Fisher et al., 2019; Gallino et al., 2017; Gallino & Moreno, 2014; Ofek et al., 2011). These strategies influence customer shopping behavior, including willingness to pay, strategic wait, and channel migration (Aviv & Pazgal, 2008; Bell et al., 2014; Cachon et al., 2005; Kök & Xu, 2011; Su & Zhang, 2008). Example strategies include communicating product availability information (Gallino & Moreno, 2014; Gao & Su, 2017a; Gao & Su, 2017b), setting up physical showrooms/pop-up stores/experiential stores (Bell et al., 2017; Zhang et al., 2019), introducing virtual fitting (Gallino & Moreno, 2018), and making online product reviews available for B&M shoppers (Gallino & Moreno, 2019).

We complement this set of research by empirically exploring demand and operational outcomes associated with providing B&M store product availability information in the omnichannel context, coupled with product assortment differentiation between the online and B&M channels. Three streams of literature that closely relate to our study are highlighted.

The first stream of literature examines customer strategic reactions when provided with inventory/product availability information through virtual communications. Though most of these studies are based on a single channel setting, they establish the causal relationship between communicating product availability information and customer shopping decisions. Specifically, Cui et al. (2019) examine the effect of revealing the percentage of inventory remaining on customer purchase decisions within the Amazon lightning deal context. They find that customers use this inventory information to update their belief on product value and that higher past sales spur subsequent customer demand. In addition, Calvo et al. (2020), Park, Rabinovich, Tang, and Yin (2020), Allon and Bassamboo (2011), and Park, Rabinovich, Tang, Yin, and Yu (2020) investigate the impact of communicating inventory scarcity messages on sales in online retailing. In particular, Calvo et al. (2020) use data obtained from an online flash-sale retailer to show that inventory scarcity information stimulates sales, but at the same time, may lead to more product returns as customers rush to make their purchase decisions. On the contrary, Park, Rabinovich, Tang, and Yin (2020) find that the sales of durable products with regular inventory replenishment cycles may be hindered by revealing low-inventory levels, as customers may interpret such information as a signal of inferior product and service quality. Another study by Park, Rabinovich, Tang, Yin, and Yu (2020) use the EOQ framework to show that with slow-moving items, the positive impact of disclosing inventory scarcity information on sales only applies to products with extreme retail margins. Lastly, Allon and Bassamboo (2011) discover that customers may view unverifiable product availability information as "cheap talk" and, as a result, would not change their buying behavior.

Our study contributes to this literature stream by extending the investigation on the impact of sharing product availability information from a single-channel to a multi-channel context. This makes our study substantially different from the extant literature for the following reasons. First, for online retailers, since the website is their only touchpoint with customers, the primary motivation for these retailers to disclose product availability information is to influence customers' value judgments of the product so as to persuade them to "buy here, buy now." On the other hand, omnichannel retailers maintain multiple touchpoints with customers and use this information as an integration tool to coordinate the products and services offered across channels and reduce shopping "frictions" from information asymmetry (Bell et al., 2014). The ultimate goal is to enhance customer experience and build customer loyalty (Gallino & Moreno, 2019). Second, with multiple touchpoints in the system, customers have more options to adjust their shopping trajectories, leading to more complex outcomes associated with this information sharing strategy.

The second stream of literature that our work builds upon explores the relationship between product assortment, customer demand, and firm operations (Van Ryzin & Mahajan, 1999; Rooderkerk & Kök, 2019; Cachon et al., 2005; Wang & Sahin, 2018; Cachon et al., 2019). Previous work has identified the linkage between product assortment and customer search behavior and concluded that a broader product assortment helps reduce customer search costs (Cachon et al., 2005; Wang & Sahin, 2018) and increases sales (Cachon et al., 2019). Rooderkerk and Kök (2019) point out that with omnichannel retailers, the most common format of cross-channel product assortment planning is "asymmetric integration," where the online channel carries a wider variety than the B&M channel. Several studies have examined strategies that address the product assortment gap between the online and offline channels. Gallino et al. (2017) explore the outcomes of an omnichannel strategy that allows customers to ship online-exclusive items for in-store evaluation ("Buy Online, Ship-to-Store"). They find that the program increases sales dispersion among SKUs as customers purchase more lower-selling products (mostly only available online) when offered in-store evaluation opportunities. Furthermore, Akturk et al. (2018) investigate the effects of the same strategy and reveal customers switching from the online channel to shop at the B&M stores following the implementation, leading to increased B&M sales and cross-channel returns. We contribute to this stream by exploring another omnichannel strategy that integrates online and offline product assortment information.

Lastly, our study also builds on the literature stream that explores customer research shopping and how it is influenced by omnichannel operations. Bell et al. (2017), Gao and Su (2017a, 2017b), and Gallino et al. (2017) evaluate the economic value of providing customers additional tools to learn about product fit and the channel switching behavior triggered by this new information. Gallino and Moreno (2014), Gallino et al. (2017), and Akturk et al. (2018) examine customer channel preference changes following the rollout of new order fulfillment options. Furthermore, Bell et al. (2017), Akturk et al. (2018), and Gallino and Moreno (2018) extend this analysis to encompass the impact of firm omnichannel operations on customer product returns behavior, including the overall return rate and cross channel return behavior. The studies above represent a broad set of literature investigating how customers are affected by the new information/service offered and adjust their subsequent shopping decisions in an omnichannel environment.

Overall, our study provides an empirical understanding of a widely adopted yet understudied omnichannel information sharing strategy in a common and generalizable retail setting. Specifically, our study seeks to answer the following three research questions: (1) what is the impact of the policy on total sales and channel-level sales, (2) how does the policy affect online order fulfillment, and (3) how does the policy affect customer returns. Moreover, by using a proprietary dataset, we contribute to the growing body of empirical research that explores the change in a customer's shopping trajectory when interacting with multiple touchpoints operated by the same retailer.

3 | RESEARCH BACKGROUND AND HYPOTHESES DEVELOPMENT

3.1 | Research setting

We partnered with a Fortune 500 consumer products manufacturer headquartered in the United States. This manufacturer owns more than 10 brands that are sold worldwide. At the point of our data collection (2017), the company had annual sales of over $5 billion. We closely worked with one division (brand) of this manufacturer that independently operates in the retail sector. This retailer specializes in housewares, toys, and apparel and was launched as a traditional B&M retailer in the 1980s, adding the online channel to its operations in the mid-1990s. It currently has annual revenue of around $400 million. Importantly, all the products sold by this retailer are exclusively designed and manufactured by its parent manufacturer firm and unavailable through other major retailers.

Our retail partner offers around 2500 SKUs. The B&M channel carries approximately 80% of the product selection compared to the online channel, and this ratio

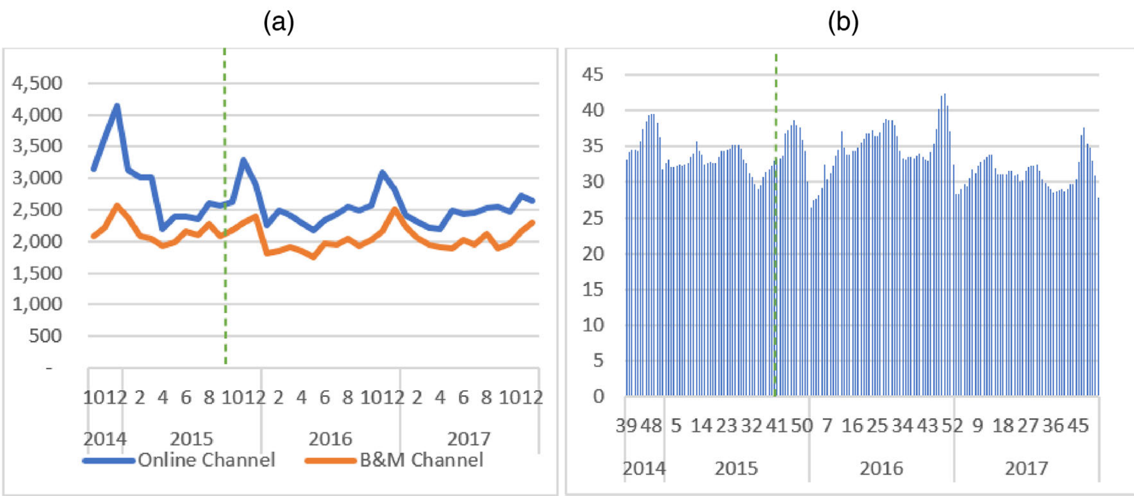

FIGURE 1 Product assortment and store inventory level. (a) Product assortment (count of SKUs) offered at the online channel versus the B&M channel. (b) Average inventory level (units) per SKU per store. The vertical dotted line identifies when the information sharing policy was launched.

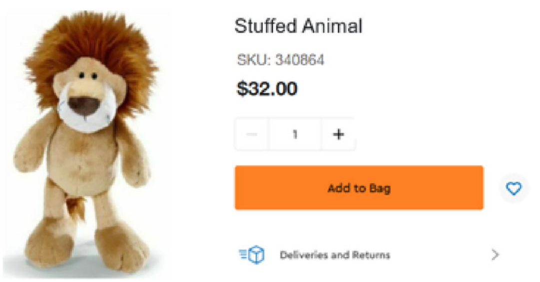

FIGURE 2 Webpage example before the implementation of the policy

remained relatively stable throughout our period of analysis[1] (Figure 1a). From a strategic standpoint, no SKU or product line is exclusively designed for a specific channel. In other words, none of the online SKUs have an "Online Exclusive" tag, and vice versa for retail store SKUs. However, each retail store carries a smaller product assortment, mainly due to fewer color/feature variations offered, limited shelf space, or a shortage in product supply from the production plant. Figure 1a demonstrates the average number of SKUs offered by the B&M channel versus the online channel over our period of analysis. Traditionally, customers did not have access to information on which SKUs were available at each retail location before physically visiting or calling the retail store (see Figure 2 for an artificial webpage example prior to the implementation of the policy).

The retailer launched the sharing retail store product availability policy for all 20 B&M stores at the same time on October 1, 2015. This policy allows a customer to obtain B&M product availability information by displaying an "Available" or "Unavailable" message after a customer selects a store location on the product webpage (see Figure 3 for an artificial webpage example after the implementation of the policy). Note that when a product is shown as "unavailable" at a specific store, it is primarily due to product assortment planning differentiation between the two channels as opposed to a temporary stock out, since our retail partner maintains a high service level at its B&M channel and carries an average daily inventory level of around 30 units per SKU per store (Figure 1b). Furthermore, since either scenario would be shown as "unavailable" on the product webpage, customers cannot identify the difference based on the information shared with them—all they know is that the product is not available at their local retail store at that time.

The retailer's business model is suitable for our study for the following reasons. First, since all products are distributed exclusively from the manufacturer to our retail partner, it helps reduce the confounding effect of competitors selling the same products. Second, our retail partner changed its product availability communication method without altering other major operational aspects. This provides us with a relatively clean research design to evaluate the effect of this information sharing strategy. Finally, our retailer maintains synchronized prices and promotional events across channels, which helps minimize the impact of pricing on customer channel switching behavior within the firm.

3.2 | Hypotheses development

To understand the impact of the policy on customer shopping decisions, we consider four possible types of

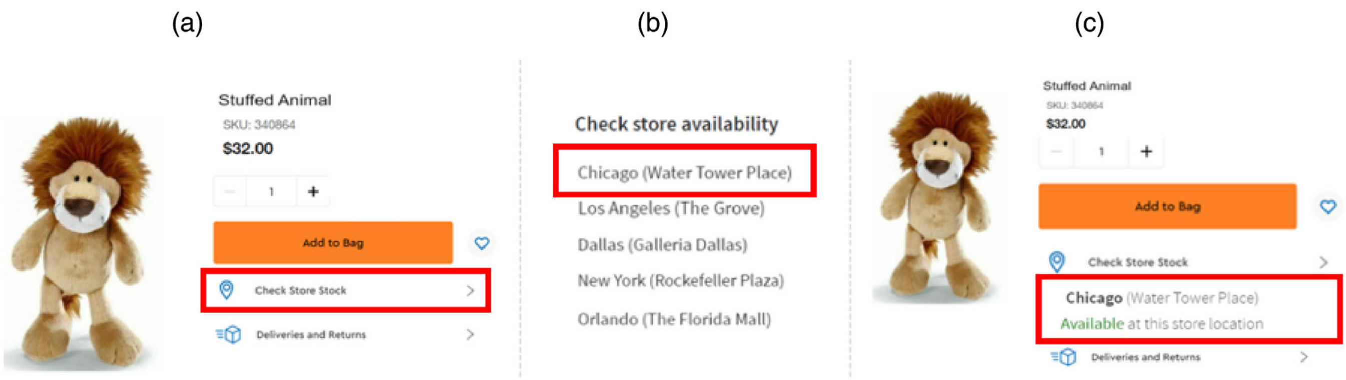

FIGURE 3 Webpage example after the implementation of the policy. (a) Webpage display after October 1, 2015 implementation. (b) Pop-up window after clicking on the "Check Store Stock" option. (c) Webpage display after selecting a specific store location.[2]

customers based on their shopping trajectory: (1) search at B&M store, purchase at B&M store (SBPB), (2) search at B&M store, purchase online ("showrooming"), (3) search online, purchase at B&M store ("webrooming"), and (4) search online, purchase online (SOPO) (Bell et al., 2014; Rooderkerk & Kök, 2019; Verhoef et al., 2007). We analyze the potential impact of the policy on these four types of customers and then predict the policy's effect on sales, order fulfillment, and product returns.

3.2.1 | Customer types and offline product availability

We begin by evaluating the effect of the policy on SBPB and showrooming customers. Disclosing in-store product availability information online should not affect SBPB customers, because these customers do not interact with the online channel (where the information is located). On the other hand, the impact of the policy on "showrooming" customers should be minor for the following three reasons. First, "showrooming" behavior primarily exists when customers compare prices and services between different retailers (Gensler et al., 2017; Rooderkerk & Kök, 2019). For example, customers may use a B&M destination to obtain tactile information and move to an e-commerce retailer for a lower price, extended warranty, etc. (Bell et al., 2014). Such pricing or service competition is minimized under our empirical setting since the products sold by our retail partner are exclusively distributed by its parent manufacturer. Furthermore, our retailer enforces synchronized pricing and promotion events across channels. Finally, even with showrooming customers, they start their search offline and would not encounter the information until they move online to complete the transactions. At that point, they should already know their selection from the B&M experience, and informing them about B&M store product availability would have little impact on their shopping decisions. Therefore, the first two types of customers are relatively unaffected by the policy.

Conversely, webrooming and SOPO customers start their research shopping online and thus can encounter this information early on during their shopping journey. Depending on the (un)availability message one encounters, a customer may choose to change their shopping type, switch between (un)available products, or move to/away from the retailer, leading to various changes in sales and operational aspects of firm performance. We will focus on the impact of the policy on these two customer types in the following analysis.

3.2.2 | The effect of sharing B&M store product availability information

We start by briefly mapping out the potential impact of the policy on webrooming and SOPO customers' shopping decisions based on the type of information they may encounter. We then develop hypotheses on the predicted overall effect of the policy on sales, order fulfillment, and product returns.

Webrooming customers

Webrooming refers to customers who start their research shopping online before moving to the B&M channel to collect additional information and complete the transaction (Gallino & Moreno, 2019). This type of customer relies on product availability at their local retail store to achieve their shopping goals. As a result, they should be the group that is most sensitive to the in-store product availability information. Specifically, when facing assured product availability, these customers will continue with their original shopping path, yet with a higher chance of completing the transaction at the store. On the other hand, webrooming customers can also encounter product unavailability information when researching online. In this case, the chance of continuing to visit the B&M channel is significantly lower because customers will not be able to purchase or inspect the item in the store. At this point, some webrooming customers may choose not to purchase from the retailer if the B&M experience is an essential component of their shopping journey. Others may switch to SOPO to obtain the product(s) marked as "Unavailable" at their local retail store.

Search online, purchase online (SOPO) customers

Unlike webrooming customers, offline shopping is not a component of SOPO customers' initial plans. Nevertheless, these customers can still encounter the information while browsing the website and may use the information to adjust their shopping plan. Specifically, SOPO customers may reconsider the utility they can obtain from the offline channel when they observe assured in-store availability information on the items they browse. Under this scenario, a SOPO customer may switch to webrooming and complete the transaction offline.

SOPO customers can also encounter product unavailability information. While some SOPO customers may not be affected by this information and continue to purchase the product online, others may be hesitant to buy the product if they are concerned that limited availability indicates low popularity, inferior quality, end of a product life cycle, etc. (Park, Rabinovich, Tang, & Yin, 2020). Thus, similar to webrooming customers, product unavailability information can also result in a "no purchase" decision for SOPO customers.

The overall effect of the policy

Based on the above analysis, we can see that depending on the (un)availability information encountered, customers may adjust their shopping trajectories within the channels offered by the retailer, or decide not to purchase from the retailer. Among these possible changes, the proportion of customers who would be completely discouraged by the in-store product unavailability information should be relatively small. This is because under an omnichannel context, B&M channel product unavailability does not mean that a customer cannot obtain the product from the retailer. Instead, they have the option of "having the store come to them" by ordering the products online and conducting the physical product attribute evaluation at home (Bell et al., 2014). Our retailer further eliminates customers' hesitation with this option by maintaining a very lenient return policy (free in-store and mail-in returns with no time limit) (Abdulla et al., 2019; Nageswaran et al., 2020; Wood, 2001). Thus, despite customers potentially moving away from the retailer when encountering B&M store product unavailability information, we anticipate that they compose a relatively small proportion of the population.

We now focus our attention on customers who adjust their shopping plans within the channels offered by the retailer. The information may cause three types of changes in SOPO and webrooming customers: (1) SOPO customers migrate to the B&M channel to take advantage of assured evaluation or fulfillment opportunities, (2) webrooming customers increase the likelihood of actual store visits with confirmed product availability, and (3) webrooming customers migrate to the online channel to avoid product unavailability at the B&M channel. The magnitude of these effects can be different because the value of the information added towards each group's shopping planning process varies. First, webrooming customers heavily rely on the product availability at their local retail store to achieve their shopping goals; therefore, they should be more sensitive to this information and actively utilize this product lookup function after its launch. On the other hand, SOPO customers do not directly rely on offline product availability to achieve their shopping goals. This can lead to a proportion of SOPO customers not even checking the availability information prior to order placement. Thus, sharing in-store product availability information online should primarily affect webrooming customers.

Second, the product availability/unavailability message can also influence webrooming customers in two ways. First, shopping in the B&M channel is already part of a webrooming customer's initial plan even without the product availability information. Although confirmed product availability messages may increase the likelihood of actual store visits, the magnitude should be relatively minor. On the other hand, product unavailability information alerts customers that their shopping goals cannot be achieved with their original plans, and adjustments are needed to avoid utility loss. Thus, we expect webrooming customers to react more strongly to product unavailability messages disclosed by the policy.

In summary, sharing in-store product availability information online should primarily impact webrooming customers' shopping trajectories, leading some to switch to SOPO after discovering that the product assortment at their local store cannot meet their expectations. We therefore hypothesize:

Hypothesis 1. Sharing brick-and-mortar product availability increases online sales.

Hypothesis 2. Sharing brick-and-mortar product availability decreases B&M sales.

At a product level, customers purposely migrate to the online channel to obtain those products that are not currently offered at their local retail store. This should lead to a change in the composition of online sales between "B&M available products" and "B&M unavailable products." In particular, we expect to see a rise in the proportion of online sales generated from products that are unavailable at B&M stores. This results in the following hypothesis:

> **Hypothesis 3.** Sharing brick-and-mortar product availability increases the proportion of online sales from B&M unavailable products.

Next, we analyze the effect of the policy on the company's overall sales. We expect the policy to positively impact the retailer's total sales for the following reasons. First, before implementing the policy, customers may start their search online, form a mental commitment to a product, and expect it to be available for further inspection and purchase at their local store. If they later find that the product is unavailable after traveling to the store, they may leave the store empty-handed or reluctantly switch to a substitute product. Both scenarios can lead to customer disappointment towards the brand, resulting in potential lost sales. This kind of situation can be mitigated after the implementation of the policy as customers are "pre-warned" about the product selection at their local retail store. Second, the policy improves the overall cross-channel product assortment transparency, which helps reduce the information asymmetry between the retailer and its customers. The new information helps customers make shopping plans more smoothly and wisely, which should help boost customers' overall satisfaction towards the retailer, leading to an expansion of overall demand (Bell et al., 2014). We therefore hypothesize:

> **Hypothesis 4.** Sharing brick-and-mortar product availability increases total sales.

Finally, customer channel migration behavior generated by the policy may also affect the fulfillment and returns aspect of firm operations. Once customers move their purchase from the B&M channel to the online channel, they lose the in-store physical evaluation opportunity and will have to judge product attributes solely based on online descriptions (Bell et al., 2014). This will potentially lead to a higher chance of purchase dissatisfaction and increased product returns (Akturk et al., 2018; Bell et al., 2017; Hong & Pavlou, 2014). At the same time, since switching to the online channel no longer allows for immediate possession of the item, customers with time-sensitive needs may choose premium shipping services to make up the lead time between placing and receiving their orders (Fisher et al., 2019). We therefore hypothesize:

> **Hypothesis 5.** Sharing brick-and-mortar product availability increases the proportion of online orders using expedited shipping.

> **Hypothesis 6.** Sharing brick-and-mortar product availability increases product returns.

4 | DATA AND METHODOLOGY

4.1 | Data description

Data was obtained from this retailer for the time period between October 2014 and December 2017, 12 months before and 27 months after implementing the policy. During this time frame, 16 B&M stores were in operation for the entire time of analysis, while another four opened during the first 11 months (before implementing the policy). None of these stores closed during the 39-month period.

We obtained point-of-sale records, online fulfillment records, daily inventory records, and product return records from this retailer. The point-of-sales dataset covers all transactions in the online and B&M channels, including over 13.78 million transactions from 37,636 ZIP codes, which is more than 90% of all U.S. ZIP codes. For each transaction, we observe the date, channel, and store where it took place, the size and value of the transaction, the stock-keeping unit (SKU) of the product sold, the price at which the SKU was sold, and the MSRP price of the SKU. Additionally, we obtained data on the online fulfillment services (standard versus expedited shipping) used for all online transactions.

We also collected daily SKU-level inventory records at each retail store for the same 39-month period. For each record, we obtained the date, location, SKU, and number of sellable units carried. This dataset serves as the key to identifying product availability at each retail store, which is the information that is disclosed to customers by the new policy. Finally, we collected data on customer product returns to trace whether the item(s) in each transaction is(are) eventually returned to the retailer.

4.2 | Sample and methodology

4.2.1 | Sample description

To examine the impact of sharing B&M product availability on overall sales and channel level sales, we aggregate the transaction-level data to a "ZIP Code—Month" level, based on the ZIP Code information associated with each order. For online orders, the ZIP Code is based on the shipping address of the order. For B&M store orders, the ZIP Code reflects the customer's household location and was collected by our retail partner at check out. We choose to focus our analysis at the ZIP Code level since it defines a relatively small geographic unit. This increases the chance that customers within the same unit share common socioeconomic characteristics. Demographic information is also available at the ZIP Code level, which allows us to identify differences across ZIP codes (Avery et al., 2012; Bell et al., 2017). Note that several temporary retail locations were in operation for a short period as a marketing campaign to promote the brand. All of these temporary locations were opened outside the trade areas of existing B&M stores. To ensure a clean research design with a single treatment, we consider only ZIP codes whose distance to the closest B&M store is unaffected by the opening and closing of retail locations. This leaves us with 10,725,888 transactions from 28,433 ZIP codes from 16 B&M stores and one online store.

For the rest of the analysis, we focus on ZIP codes that generated at least one order each month between October 2014 and December 2017. This criterion ensures that our analysis is not biased by those ZIP codes that have extremely low demand over time and helps estimate the impact of the policy where it is most relevant (Bell et al., 2017). It also helps avoid a large number of observations with zero sales when conducting analysis at a weekly level. 5500 out of 28,433 ZIP codes meet this criterion and, in turn, generate 5,950,139 transactions, accounting for 57.8% of the total sales generated by all ZIP codes that are unaffected by store openings and closings. Additional sample selection methods are examined in the robustness check section.

4.2.2 | Methodology and research design

We employ the difference-in-differences (DiD) approach, a method widely adopted in the retail operations literature to study the impact of a policy intervention using a quasi-field experimental design (Akturk et al., 2018; Avery et al., 2012; Bell et al., 2017; Fisher et al., 2020; Gallino et al., 2017; Gallino & Moreno, 2014). Angrist and Pischke (2008) summarize the major merits associated with this method. In general, the DiD approach allows both a "before and after" comparison within a population group affected by the intervention and a contemporary comparison across population groups that are affected/unaffected by the intervention. This helps rule out the impact of other factors that may affect sales but are unrelated to the intervention of interest (e.g., new product launches, end-of-season sales, etc.).

We assign customers to the "Treatment Group" and the "Control Group" based on their household locations. Specifically, we use the distance between a customer's household ZIP Code and the nearest retail store to measure whether the policy is relevant to a customer. The underlying rationale is that the further a customer is located from a retail store, the greater the time and monetary costs of the trip and, consequently, the less likely it would be for a customer to visit a store. Customers who are a sufficient distance from a store are, therefore, relatively "immune" from changes that take place at the retail store (Akturk et al., 2018; Bell et al., 2017; Gallino & Moreno, 2014). A 50-mile radius is adopted to define a B&M store's trade area, and customers who live within the 50-mile radius of a B&M store are assigned to the treatment group, while customers who live outside the 50-mile radius of a B&M store are assigned to the control group. We choose this value as a starting point since it aligns with our retail partner's recognition of the trade area of a B&M establishment and is consistent with prior research in the omnichannel domain (Akturk et al., 2018; Gallino & Moreno, 2014). Based on this criterion, 2825 ZIP codes out of 5500 ZIP codes are assigned to the treatment group, while the remaining 2675 are assigned to the control group.

A necessary discussion when using the DiD approach in a quasi-field experiment setting is to ensure that the treatment and control groups are comparable (Akturk et al., 2018; Avery et al., 2012; Bell et al., 2017; Gallino & Moreno, 2014). In an ideal setting, customers should be randomly assigned to the treatment group and the control group, and the only difference between the two groups is the intervention of interest. However, retail store locations are not randomly selected in our retail setting. To address the endogeneity concern, we assess the plausibility of the parallel trend assumption by looking for supportive evidence that can suggest it is acceptable to assume that the "treatment effects" are not primarily driven by the initial trends difference between the treatment and control groups before the policy rollout (Bronnenberg et al., 2020; Gallino & Moreno, 2014; Huntington-Klein, 2021). Note that the parallel trends are "inherently unobservable," and one can provide suggestive evidence at best (Bronnenberg et al., 2020; Huntington-Klein, 2021). We use the following methods to address this concern. First, we compare the pre-implementation sales trend from the treatment group to the same-time-period sales trend from the control group. Second, we conduct a *placebo test* by creating "fake" policy rollout dates in the middle of the pre-implementation period and estimate its effect on customer demand.

TABLE 1 Demographic statistics before and after matching

	Before matching			After matching		
	Treatment	Control	Z score	Treatment	Control	Z score
Population	29,859	23,933	−13.7	26,786	26,611	−0.3
Median household income	91,628	67,670	−32.6	73,458	75,602	3.2
Total retail sales (millions of $)	542	448	−7.1	483	495	0.7
Unemployment rate (%)	4.7	5.0	7.6	4.9	4.9	−1.2
Number of ZIP codes	2825	2675		1644	1644	

TABLE 2 Variable descriptions

Variable name	Definition
Dependent variables	
Online sales$_{it}$	Dollar sales at the online store coming from ZIP code i in month t
B&M sales$_{it}$	Dollar sales at the B&M stores coming from ZIP code i in month t
Total sales$_{it}$	Dollar sales at the online store and B&M stores coming from ZIP code i in month t
Percentage of online sales from B&M unavailable products$_{it}$	Proportion of online sales coming from ZIP code i in month t generated by products not concurrently offered at a customer's closest retail store
Percentage of online orders using expedited shipping$_{it}$	Proportion of online orders with expedited shipping service from ZIP code i in month t
Return rate$_{it}$	Proportion of online and B&M transactions returned from ZIP code i in month t
Independent variables	
Treatment group$_i$	Dummy variable that equals 1 if ZIP code i is within the trade area of a B&M store
Policy$_t$	Dummy variable that equals 1 for months after October 2015
Treatment group$_i$ × policy$_t$	Interaction term of two dummy variables that only equals 1 if ZIP Code i is affected by the implementation of the policy
Control variables	
Monthly dummies$_t$	Dummy variable for each year-month t
ZIP code dummies$_i$	Dummy variable for each ZIP code i

Detailed analysis is presented in Appendix B. Overall, we do not find evidence to support that our findings are primarily driven by pre-existing sales trends.

Furthermore, we use propensity score matching to construct a sample where the treatment group shares similar demographic characteristics with the control group (Akturk et al., 2018; Bell et al., 2017; Gallino & Moreno, 2014; Imbens & Wooldridge, 2009; Rosenbaum & Rubin, 1983). This helps increase the plausibility of the parallel trend assumption holding and allows for a more precise estimate of the policy effects (Huntington-Klein, 2021). We first estimate the propensity score for each ZIP Code in our sample using a logit model. Specifically, we collect ZIP-Code level demographic information from the *SimplyAnalytics*[3] database and match ZIP codes within the treatment group to those within the control group based on four demographic variables: population, median household income, total retail sales, and unemployment rate, which align with our retailer partner's evaluation of the size and market potential for a market region (Bell et al., 2017). We use one-to-one nearest neighbor matching with a caliper value of 0.3. From the treatment group, 1644 of the 2825 ZIP codes have a suitable control group match. The matching significantly reduced the difference between the two groups of customers Table 1 demonstrates the key demographic comparisons between the treatment and control groups before and after matching. Our final sample includes 128,232 "ZIP Code-Month" observations from 3288 ZIP codes over 39 months. Additional sample selection methods are examined in the robustness check section.

TABLE 3 Summary statistics

Variable	Obs.	Mean	SD	Min	Max
Dependent variables					
B&M sales	128,232	1089	1307	0	37,172
Online sales	128,232	1773	3188	0	236,717
Total sales	128,232	2746	3769	0.86	240,270
Percentage of online sales from B&M unavailable products	123,623	30.16	22.38	0	100
Percentage of online orders using expedited shipping	123,623	9.16	15.60	0	100
Return rate	128,232	2.36	4.80	0	100
Independent variables					
Treatment group	128,232	0.50	0.50	0	1
Policy	128,232	0.69	0.46	0	1
Treatment group × policy	128,232	0.35	0.48	0	1

Note: The number of observations is slightly smaller for *percentage of online sales from B&M unavailable products* and *percentage of online orders using expedited shipping* because some ZIP codes have zero *Online Sales*.

4.2.3 | Variables and model

In this section, we provide descriptions of the variables and model used to test the hypotheses raised in Section 3. Table 2 includes the definitions of all variables. Table 3 presents summary statistics.

Dependent variables

To examine the impact of the policy on customer demand in the online and B&M channels, we construct *Online Sales* and *B&M Sales*, by summing all online and B&M transactions, to a "ZIP Code—Month" level, respectively. In addition, we also calculate *Total Sales* by summing *Online Sales* and *B&M Sales*. We use the inventory record dataset to check whether each item sold online is concurrently available at the customer's local (closest) retail store, which is the information made available to customers by the policy. This allows us to categorize online sales into "sales from products available at the local store" and "sales from products unavailable at the local store." We only focus on whether a product is in stock at the closest store because the distance between any two stores is over 200 miles, making it unrealistic for customers to travel to a "second closest" retail store. The variable is calculated as follows:

$$\text{Percentage of online sales from B\&M unavailable products} = \frac{\text{Online sales from B\&M unavailable products}}{\text{Total online sales}}$$

We also construct the *percentage of online orders using expedited shipping* variable by calculating the percentage of online transactions that used "expedited shipping" service as opposed to the "standard shipping" option. The expedited shipping option is usually \$10–\$20 more expensive than the standard shipping option, but the speed difference is considerable. "Standard shipping" takes an average of 6–10 business days, while "expedited shipping" requires an average of 2–3 business days. Finally, we aggregate the number of return transactions to the "ZIP Code—Month" level based on the original transaction's date and household ZIP Code and divide it by the total number of transactions during the same period to calculate the *Return Rate*.

Independent variables

The construction of the treatment group and the control group naturally allows for the use of an indicator variable, *Treatment Group*, which equals one if a ZIP Code is within the trade area of a retail store and zero otherwise. We also construct the *Policy* variable to differentiate the time period before and after the rollout of the sharing retail store product availability program. This indicator variable equals zero for the period between October 1, 2014, and September 30, 2015, and equals one for the period between October 1, 2015, and December 31, 2017.

Control variables
We include fixed effects for both the ZIP Code and the time dimensions of our panel dataset, which is consistent with a large number of retail operation papers (Bell et al., 2017; Fisher et al., 2019; Gallino et al., 2017; Gallino & Moreno, 2014). Specifically, we include one dummy variable for each ZIP Code and one dummy variable for each year-month.

Model
Building on the difference-in-differences approach, we use the following model to evaluate the impact of the policy:

$$\mathrm{DV}_{it} = \alpha_1 \text{Treatment group}_i \times \text{policy}_t + \mu_i + W_t + \varepsilon_{it} \quad (1)$$

where the DV represents each dependent variable discussed earlier in this section. Standard errors are clustered around each ZIP Code and time period to address potential heterogeneity in the treatment effects between ZIP codes and across time (Abadie et al., 2017; Velentgas et al., 2013). Our variable of interest, *Treatment Group * Policy*, is a dummy variable that accounts for the impact of the policy on the dependent variable from ZIP codes located within the trade area of a retail store (*Treatment Group* = 1) following the rollout of the policy (*Policy* = 1). To fully control for the variations that are unrelated to the rollout of the policy, we include both ZIP Code fixed effects μ_i (for each of the 3288 ZIP codes) and time fixed effects W_t (for each of the 39 months).

5 | RESULTS AND DISCUSSION

Based on the above model, we empirically test the hypotheses raised in Section 3 regarding the impact of sharing retail store product availability information on customer demand (H1, H2, H3, and H4), online order fulfillment (H5), and product returns (H6).

5.1 | Impact on customer demand

We begin our analysis by examining the impact of sharing retail store product availability information on sales at the channel level. The following regression specification is adopted:

$$\log(\mathrm{DV})_{it} = \beta_1 \text{Treatment group}_i \times \text{policy}_t + \mu_i + W_t + \varepsilon_{it} \quad (2)$$

where the dependent variable is *Online Sales* and *B&M Sales* from ZIP Code i during month t for Model (1) and Model (2), respectively. A logarithmic transformation is performed to correct the right-skewed distribution of the variable and to ease interpretation. Note that the sample is slightly smaller at the channel level because certain "ZIP Code-Month" observations generate zero online or B&M sales and are therefore dropped when performing the logarithmic transformation on the dependent variable. We show the consistency of our results with these observations included in the robustness check section.

Results are presented in Table 4, Columns 1 and 2. We observe that the policy exerts a positive and significant effect on online sales and a negative and significant effect on B&M sales, providing support for H1 and H2. Specifically, the trade area in the vicinity of a retail store experiences an average increase in online sales of 13.1%[4] (95% confidence interval [6.8%, 19.8%]) and a decrease in B&M sales of 2.6% (95% confidence interval [−4.6%, −0.5%]). The results reveal a distinct impact of the policy at the channel level and provide the first evidence that the policy may cause some customers to migrate from the B&M channel to the online channel after encountering product unavailability information.

To further unfold the channel level mechanism, we investigate the composition of online sales to see whether customers are attracted to the online channel specifically to obtain those products that are not concurrently offered at their local retail store. To do so, we examine the impact of the policy on online sales from B&M unavailable products using the following equation:

$$\text{Percentage of online sales from B\&M unavailable products}_{it} = \gamma_1 \text{Treatment group}_i \times \text{policy}_t + \mu_i + W_t + \varepsilon_{it} \quad (3)$$

Results are presented in Table 4, Column 3. The *Percentage of online sales from B&M unavailable products* experiences a positive and significant increase of 4.4 percentage points (95% confidence interval [3.7 percentage points, 5.2 percentage points]) from customers who live within the trade area of a retail store after the launch of the policy. On average, 30.2 percentage points of the online sales are generated from products that are not

TABLE 4 The impact of the policy on sales

	log (Online sales) (1)	log (B&M sales) (2)	Percentage of online sales from B&M unavailable products (3)	log (Total sales) (4)
Treatment group × policy	0.123***	−0.026*	4.435***	0.034***
	(0.028)	(0.011)	(0.381)	(0.008)
Fixed effects	Yes	Yes	Yes	Yes
Time effects	Yes	Yes	Yes	Yes
Propensity score matched sample	Yes	Yes	Yes	Yes
N	123,623	121,945	123,623	128,232
R square	0.474	0.216	0.156	0.432

Note: Clustered standard errors in parentheses. $^*p < .05$; $^{**}p < .01$; $^{***}p < .001$.

offered at a customer's local retail store, and a 4.4 percentage point increase represents a 14.6% (4.4/30.2) change in the dependent variable, surpassing the 13.1% change in overall online sales. This finding provides support for H3 and reveals further evidence that the product assortment gap contributes to customer channel migration behavior following the implementation of the policy.

Next, we evaluate the impact of the policy on the retailer's overall sales. We use the same regression specification as Equation (2), except the dependent variable is *Total sales*.

$$\log(\text{Total sales})_{it} = \theta_1 \text{Treatment group}_i \times \text{policy}_t + \mu_i + W_t + \varepsilon_{it} \tag{4}$$

Results from Model (4) are listed in Table 4, Column 4, and indicate a positive and significant net effect of the policy on total sales, providing support for H4. Specifically, the policy is associated with an average increase in total sales of 3.5% (95% confidence interval [1.9%, 5.1%]) for those ZIP codes within the trade area of a retail store relative to those ZIP codes outside the trade area of a retail store. This finding, together with the results discovered at the channel level, suggests that sharing retail store product availability upfront helps customers proactively adjust their shopping plans to avoid unsuccessful shopping trips, leading to fewer lost sales and enhanced customer satisfaction.

5.2 | Impact on order fulfillment and product returns

Whereas the above analysis tested the impact of the policy on sales, we now turn our attention to exploring whether this information sharing strategy also affects order fulfillment and product returns. In particular, for customers who want to touch and feel the product, their uncertainty cannot be resolved without the offline, physical experience. Customers that switch to the online channel without being able to physically inspect the product are more likely to be unsatisfied with their purchase and initiate product returns. Additionally, customers with time-sensitive demand who switch to the online channel are more likely to select the expedited shipping service.

TABLE 5 The impact of the policy on order fulfillment and product returns

	Percentage of online orders using expedited shipping (5)	Return rate (6)
Treatment group × policy	0.348*	0.150*
	(0.174)	(0.060)
Fixed effects	Yes	Yes
Time effects	Yes	Yes
Propensity score matched sample	Yes	Yes
N	123,623	128,232
R square	0.126	0.010

Note: Clustered standard errors in parentheses. $^*p < .05$; $^{**}p < .01$; $^{***}p < .001$.

To test the above two hypotheses, we use the same sample as our previous analysis. The regression specification is identical to Equation (1), except that the dependent variables are the *percentage of online orders using expedited shipping* for Model (5) and *return rate* for Model (6).

Results are presented in Table 5. Column 1 demonstrates the impact of sharing retail store product availability on the percentage of online orders using expedited shipping, and Column 2 demonstrates the impact of the policy on the overall product return rate. We find that following the rollout of the policy, the percentage of online orders that chose to use expedited shipping increased by 0.4 percentage points (95% confidence interval [0.0 percentage points, 0.7 percentage points]), providing support for H5. Although the magnitude may seem small, note that, on average, only 9.2 percentage points of online orders select expedited shipping, and a 0.4 percentage point increase represents a 4.3% (0.4/9.2) change in the dependent variable. This finding provides additional evidence that the change in sales at the channel level is partially driven by webrooming customers with time-sensitive needs switching to SOPO after finding out that the local retail store does not carry the item.

Regarding product returns, the policy leads to an average increase in the return rate of 0.15 percentage points (95% confidence interval [0.0 percentage points, 0.3 percentage points]), which supports H6. The average return rate at the firm level during our period of analysis is 2.4 percentage points, and a 0.15 percentage point increase represents a 6.3% (0.15/2.4) change in the dependent variable. This finding indicates that the policy attracts some customers who originally planned to visit the B&M channel for additional information but switched to the online channel after discovering that their item is unavailable at their local retail store. The inability to observe the product before purchase leads to an increase in the return rate.

6 | ROBUSTNESS CHECKS

We conduct several robustness checks to ensure that our results are not driven by sample construction, data aggregation, trade area definition, regression estimation method, extreme transaction values, the inclusion/exclusion of demographic control variables, and matching criteria. These additional analyses confirm that our findings are reliable. Results tables are presented in Appendix C.

6.1 | Alternative sample constructions

We first examine whether our results are sensitive to the way the sample is constructed. To do so, we adopt the following alternative sample selection criteria: (1) ZIP codes that consecutively generate at least two orders each month as compared to our criteria of one order per month, and (2) ZIP codes that generate a total of 169 orders or more during our 39-month (169 weeks) period of analysis (on average one order per week). Alternative (1) produces a sample size of 4245 ZIP codes (2598 ZIP codes after matching). Alternative (2) produces a sample size of 8364 ZIP codes (4560 ZIP codes after matching). Note that alternative (2) includes a larger number of ZIP codes because we do not require each ZIP Code to generate consecutive orders each week, but, *on average*, one order per week. Results based on both samples show consistency with our main findings and are presented in Tables C1 and C2.

6.2 | Alternative data aggregation method

Second, we investigate whether our findings can be replicated with more granular level data. To do so, we aggregate our data at a weekly level instead of a monthly level and reproduce our main analysis. Note that there are only 620 ZIP codes that generate *consecutive* orders every week throughout our period of analysis. To ensure a reasonable sample size, we focus on ZIP codes that, *on average*, generate one order each week (a total of 169 orders or more). This leaves us with 4560 ZIP codes after performing propensity score matching (2280 ZIP codes in the treatment group and 2280 ZIP codes in the control group). Results are presented in Table C3 and demonstrate consistency.

6.3 | Alternative regression estimation method

Third, we test the robustness of our results using a different regression estimation method. In our original model, we dropped a small proportion of our sample to perform a logarithmic transformation on the dependent variable. This leads to a minor difference in the number of observations across models. The dropped units may be disproportionately distributed between the treatment group and the control group, so it is necessary to examine whether our findings are affected by these observations.

We tackle the above concern by adopting a fixed-effects Poisson regression to replicate our analysis. The fixed-effects Poisson regression is useful in handling non-negative but skewed data and has been adopted in previous literature to address similar data concerns (Bell et al., 2017; Wang & Goldfarb, 2017). Results presented in Table C4 indicate that our findings are not biased by the dropped observations or regression estimation method.

6.4 | Alternative trade area definitions

We also test whether our results are sensitive to the definition of the trade area. As previously mentioned, both a

50-mile radius and a 30-mile radius have been widely adopted in the extant literature to define the trade area of a B&M store (Akturk et al., 2018; Bell et al., 2017; Gallino & Moreno, 2014). Our original analysis uses a 50-mile radius to define the trade area. In this section, we also run our analysis using a 10-mile, 20-mile, 30-mile, 40-mile, and 60-mile radius to define the trade area. Note that using different cutoff values will result in significantly different sample sizes after propensity score matching because the number of ZIP codes that fall into the treatment group versus the control group varies depending on this definition. For example, when using a 10-mile radius trade area definition, 638 ZIP codes out of the 5500 ZIP codes are assigned to the treatment group, while 4862 ZIP codes are assigned to the control group. Out of the 638 treatment group ZIP codes, 637 have a suitable match in the control group, leading to a sample size of 1274 ZIP codes. As the trade area radius gets larger, the number of ZIP codes assigned to the treatment group increases while the number assigned to the control group decreases, resulting in a more balanced sample and thus an increase in sample size after the matching procedure. As we get to a 60-mile trade area definition, 3057 ZIP codes are assigned to the treatment group, while 2443 ZIP codes are assigned to the control group. Out of the 3057 treatment group ZIP codes, 1657 have a suitable match in the control group, leading to a sample size of 3314 ZIP codes. Results based on each trade area definition are presented in Tables C5–C9, respectively. Figure C1 presents a graphical display of the treatment effect for each dependent variable across different trade area definitions.

6.5 | Possible impact of extreme transaction values

Fifth, we analyze whether our findings could be driven by extreme transaction values. Transactions with extreme values only occur occasionally and do not represent the regular purchasing pattern of customers. To rule out the possibility that our results are influenced by these extreme values, we winsorize transaction values by the top and bottom 0.1% (20,892 transactions winsorized), top and bottom 1% (214,183 transactions winsorized), and top and bottom 5% (1,048,733 transactions winsorized), respectively. All results remain consistent with our main findings. Tables C10–C12 present the results.

6.6 | Alternative control variables

We also examine whether our findings are robust when we directly include demographic variables used in sample matching as control variables in our regression analysis. Specifically, we match the ZIP codes in the treatment group with those in the control group based on four dimensions (population, median household income, unemployment rate, and total retail sales), which are key determinants used by our retail partner when making market expansion decisions. Our results remain consistent when these variables are included in the regression analysis as control variables. Results are presented in Table C13.

6.7 | Alternative propensity score matching criteria

Finally, we conduct robustness analysis to investigate whether our findings are sensitive to the caliper value selected when performing the propensity score matching. There exists a tradeoff between a narrow caliper versus a wide caliper. While a narrow caliper ensures a higher level of similarity between the treatment and control groups, it may also lead to an excessively high removal of observations from the sample (Austin, 2011). We adopt a caliper of 0.3 in our main analysis. We consider two additional caliper values, 0.5 and 0.05, to verify whether our findings are sensitive to this value. Table C14 reports the demographic variable comparisons between the treatment and control groups under the two caliper values. Tables C15 and C16 present our main analysis with the new caliper values and demonstrate consistency.

7 | CONCLUSION

This study examines the impact of sharing retail store product availability information on sales, order fulfillment, and product returns in an omnichannel context. We find that customers utilize this new information and adjust their shopping decisions to avoid unsuccessful shopping trips, resulting in a significant increase in online sales, but a slight decrease in B&M sales. Overall, the cross-channel product assortment transparency allows customers to adjust their shopping plans promptly and wisely, leading to fewer lost sales and less dissatisfaction towards the brand. This contributes to an increase in overall sales for the firm.

On the operational front, the policy leads to an increase in the proportion of customers using expedited shipping when shopping online. In addition, the policy induces a slight increase in product return rate. This is because when customers encounter product unavailability information, some of them switch from “webrooming” to “search online, purchase online” (SOPO). However, as the online channel is inherently associated with a delay in order fulfillment and fewer product judgment opportunities, customers adopt pro-active (fast shipping services) and post-active (product return services) methods to meet their needs.

To the best of our knowledge, this paper is the first empirical study to evaluate the impact of sharing retail store product availability information within the omnichannel context. By answering three fundamental research questions in the retail operations domain—(1) whether a company benefits from the new strategy, (2) which channel(s) benefit from the new strategy, and (3) how the new strategy affects other aspects of firm operations—our findings contribute to the ongoing discussion regarding the value and costs associated with omnichannel information integration strategies. Furthermore, by focusing on the product assortment gap between B&M locations and the online channel of a firm, we add insight to the research stream on the role and function of B&M retailing within the evolving omnichannel environment.

There are several limitations of our study that can be addressed by future research. First, we use the inventory records provided by our retail partner to measure product availability at each retail store. Doing so relies on two assumptions: (1) inventory records are accurate and (2) shelf replenishments from the storage room are prompt. These inventory records can occasionally include errors from damage, theft, and inaccuracy within the system (DeHoratius & Raman, 2008). Future research can address this issue by working with retailers who adopt technologies that report real-time inventory on the shelf.

Second, our research findings apply to products where customers are likely to utilize the new information prior to their visit. This may not be the case for all retailers or all product categories. For certain types of retailers, such as grocery stores, customers may be less interested in checking the availability information for each item on their shopping list prior to a store visit. Another example may be retailers whose superior B&M customer service is an important component of their product value (e.g., high-end jewelry retailers). In this case, informing customers about B&M product (un)availability may lead to them canceling their shopping plans rather than switching to the online channel. Future research should explore the impact of this information sharing strategy on different retail formats and different product categories.

Lastly, we use a quasi-field experimental design to study the change in sales and operational outcomes caused by implementing an information sharing strategy. A quasi-field experimental design maintains a delicate balance between internal and external validity and thus has been widely adopted in the retail operations literature (Akturk et al., 2018; Avery et al., 2012; Bell et al., 2017; Fisher et al., 2019; Forman et al., 2009; Gallino & Moreno, 2014; Gallino et al., 2017). Often, data can be directly obtained from a company's past operations, reducing the considerable time and monetary costs of conducting a field experiment. Nevertheless, it is necessary to acknowledge the value associated with randomized-field experiments, and the method should be considered when possible (Gallino et al. 2017; Goic & Olivares, 2019; Fisher et al., 2020).

The findings of our study provide several implications for industry practitioners. First, our analysis highlights the benefit of improving information transparency between a firm and its customers. The omnichannel era has allowed customers to obtain information from various sources, including offline sites, online sites, mobile applications, and social media. Therefore, companies can no longer rely on information asymmetry to take advantage of the market; instead, they should strive to provide the information customers need to make their shopping experience smoother and more enjoyable.

Second, our study sheds light on the omnichannel product assortment planning discussion and helps practitioners re-think the role of their B&M stores. Practically, it is challenging for B&M stores to carry as wide of a selection of products as the online channel due to space limitations, costs associated with inventory and shelf replenishment, and the advantages of inventory pooling (Gallino et al., 2017). With the trend of retail store size reductions, this product assortment gap will become more significant over time (Caro et al., 2020; Mou et al., 2018). Thus, companies should consider bringing additional value-added functions to the B&M channel to further distinguish its services from those offered in the online channel. This also aligns with some major retailers' recent moves, including the opening of service stores by Nordstrom that focus on tailoring and styling services (Nordstrom Press Release, 2019). At the same time, retailers should keep in mind that the product assortment difference between the two channels may affect the outcomes of omnichannel strategies. Future research should extend the investigation into what a company should offer at its B&M channel relative to the online channel in order to improve the utility customers receive from the brand.

Finally, our analysis shows that the impact of an omnichannel strategy can be complex, and practitioners should comprehensively evaluate its outcomes using measurements that can reflect both demand and operational perspectives. In our study, we find that although sharing B&M store product availability information leads to an increase in overall sales, it also increases the return rate as customers migrate from the B&M channel to the online channel. This leads one to consider how to strengthen the product fit information delivery mechanism in the online channel. Future research should continue exploring technology applications and omnichannel strategies that can help customers understand their purchases better.

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ENDNOTES

[1] On average, the B&M channel carried 77% of the online product selection before the policy implementation and 81% after the policy implementation. A Z-test confirms an insignificant difference between the two values (Z-score $= -1.3$).

[2] There are four possible webpage outcomes – Available/Unavailable at the B&M store and Available/Unavailable at the online store. We display one of the outcomes in Figure 3(c). The other three possible outcomes are included in Appendix A.

[3] SimplyAnalytics is a database that provides demographic, business, employment, consumer spending, and marketing variables at different census-tract levels. We obtained ZIP code level variables from a university-subscribed version of this database. The database can be accessed at https://simplyanalytics.com/.

[4] Since we perform a logarithmic transformation on the dependent variable, we translate the regression coefficients using the following function $(\exp(\alpha_1) - 1) \times 100\%$. This applies to the rest of the models as well.

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APPENDIX A

A.1 | Artificial webpage example

See Figure A1.

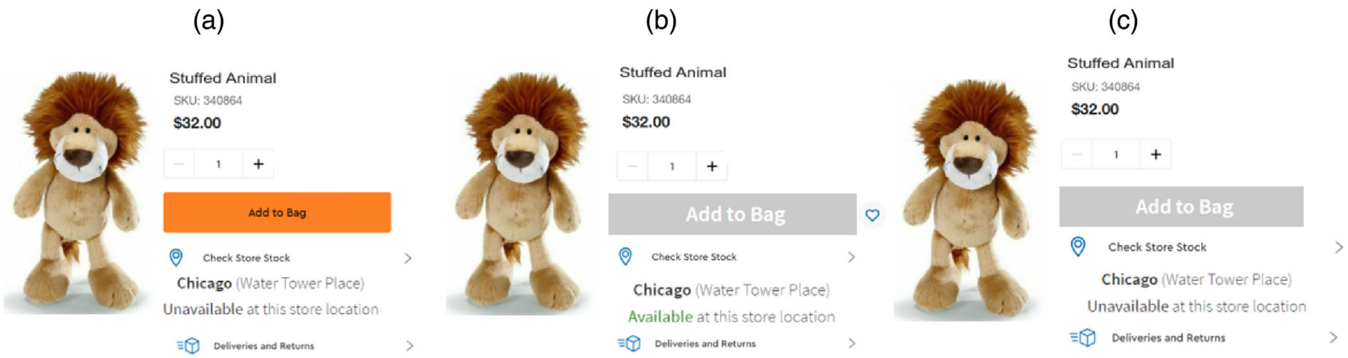

FIGURE A1 (a) Available at the online store but unavailable at the B&M store. (b) Available at the B&M store but unavailable at the online store (gray color "Add to Bag" button indicates that the item cannot be added to the shopping cart). (c) Unavailable at the online and the B&M store.

APPENDIX B

B.1 | Parallel trend test

We compare the "pre-intervention" sales trends between the treatment group and the control group to look for suggestive evidence that the sales from the treatment group were not already on a different trend prior to the rollout of the policy. We follow the methodology described in Huntington-Klein (2021) and Bronnenberg et al. (2020) and use the same set of ZIP codes from our main analysis to compare the pre-intervention sales trends for the treatment group and control group between October 1, 2014 and September 30, 2015.

$$\mathrm{DV}_{\mathrm{it}} = \xi_0 \mathrm{Trend}_t + \xi_1 \mathrm{Treatment\ group}_i{}^{*}\mathrm{Trend}_t + \mu_i + \varepsilon_{it} \tag{B1}$$

where the DV represents *log(Online Sales)*, *log(B&M Sales)*, *log(Total Sales)*, *Percentage of Online Sales from B&M Unavailable Products*, *Percentage of Online Orders using Expedited Shipping*, and *Return Rates*, respectively. $Trend_t$ denotes the count of weeks since October 1, 2014. $Treatment\ Group_i$ refers to those ZIP codes located within the trade area of B&M stores that will be affected by the policy later in time. We also include ZIP Code fixed-effects μ_i in the regression. Results are presented in

TABLE B1 Parallel trend test

	Log (online sales)	Log (B&M sales)	Log (total sales)	Percentage of online sales from B&M unavailable products	Percentage of online orders using expedited shipping	Return rate
Trend	−0.0347***	−0.0081***	−0.0240***	−0.1363	0.0754***	−0.0130
	(0.0003)	(0.0004)	(0.0002)	(0.0997)	(0.0048)	(0.0161)
Treatment group × trend	0.0004	−0.0007	−0.0003	0.0812	−0.0146	−0.0074
	(0.0004)	(0.0005)	(0.0003)	(0.0842)	(0.0081)	(0.0047)
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Time effects	No	No	No	No	No	No
Propensity score matched sample	Yes	Yes	Yes	Yes	Yes	Yes
N	38,172	37,987	39,456	38,172	38,172	39,456
R square	0.146	0.115	0.127	0.021	0.058	0.003

Note: Clustered standard errors in parentheses. $^{*}p < .05$; $^{**}p < .01$; $^{***}p < .001$.

Table B1. Note that since the coefficients of some variables have very small values, we report all results in Appendix B to four decimals. In all cases, ξ_1 is statistically insignificant ($p > .1$). This provides the first suggestive evidence that the treatment and control groups may not be on different trends before the policy implementation.

We also examine the magnitude of ξ_1 for each dependent variable despite them being statistically insignificant. By doing so, we want to further rule out the concern that the "treatment effects" we find in our main analysis are driven by the difference in pre-existing trends between the treatment and control groups. Because we did not include the time-fixed effect in the model, ξ_0 picks up the retail seasonality that applies to both the treatment and control groups. ξ_1 reveals the unique trend that may only exist within the treatment group. We multiply ξ_1 with the number of months after the policy implementation (27 months) and compare the value with the treatment effect we find for each dependent variable. The calculation and comparison are shown as follows.

For online sales, the pre-existing trend of the treatment group is 0.04% different than the control group ($\xi_1 = 0.0004$, $p > .1$). After 27 months, this trend may cause the treatment group to have 1.08% (0.04*27) higher online sales than the control group, which is much smaller than the treatment effect from the policy implementation (13.1%). For B&M sales, the pre-existing trend of the treatment group is 0.07% different than the control group ($\xi_1 = -0.0007$, $p > .1$). This trend may cause the treatment group to have a 1.89% (0.07*27) lower B&M sales than the control group after 27 months, which is smaller than the treatment effect from the policy implementation (2.6%).

We use the same method to compare the changes caused by the possible pre-existing trend in the treatment group for total sales, percentage of online sales from B&M unavailable products, percentage of online orders using expedited shipping, and return rate. For total sales, the potential decrease in total sales caused by the possible trend is 0.81% (0.03*27) compared to the policy treatment effect of a 3.5% increase. For the percentage of online sales from B&M unavailable products, the possible pre-existing trend would cause the treatment group to have a 2.19 percentage point (0.0812*27) higher outcome than the control group, which is significantly smaller than the change caused by the policy implementation (4.44 percentage points). For the percentage of online orders using expedited shipping, the treatment group's possible trend

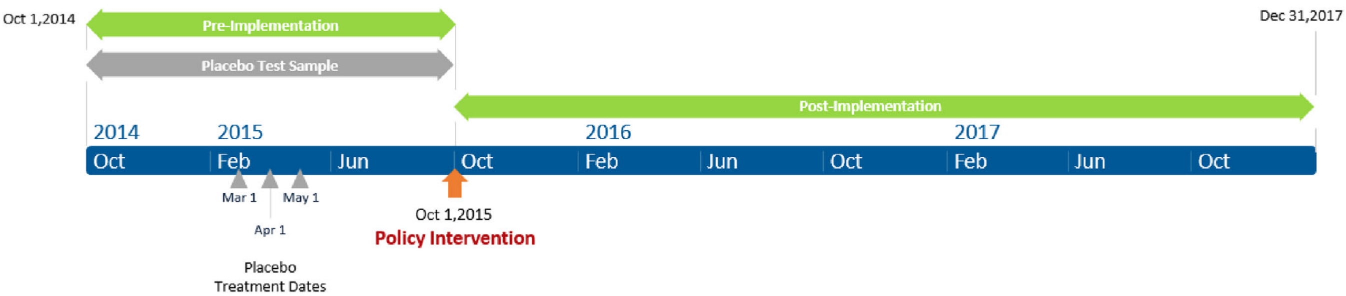

FIGURE B1 Event timeline

TABLE B2 Placebo test (March 1, 2015)

	Log (online sales)	Log (B&M sales)	Log (total sales)	Percentage of online sales from B&M unavailable products	Percentage of online orders using expedited shipping	Return rate
Treatment group × policy	0.0520	−0.0459	0.0055	0.0490	−0.0851	−0.0972
	(0.0417)	(0.0955)	(0.0069)	(0.0282)	(0.3340)	(0.1254)
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Time effects	Yes	Yes	Yes	Yes	Yes	Yes
Propensity score matched sample	Yes	Yes	Yes	Yes	Yes	Yes
N	38,172	37,987	39,456	38,172	38,172	39,456
R square	0.202	0.104	0.163	0.116	0.078	0.023

Note: Clustered standard errors in parentheses. *$p < .05$; **$p < .01$; ***$p < .001$.

may cause a 0.39 percentage point (0.0146*27) decrease in the dependent variable, which is significantly different from our main findings (0.35 percentage points increase in this variable). Lastly, the possible pre-existing trend in customer return rate is 0.0074 percentage points ($\xi_1 = -0.0074$, $p > .1$). After 27 months, this trend may cause the treatment group to have a 0.20 percentage point (0.0074*27) lower return rate than the control group. This is the opposite direction from the treatment effect we find with the policy implementation (an increase in the return rate of 0.15 percentage points). Based on this analysis, we can further clarify that the treatment effect we find in this study is not merely driven by the pre-existing trends associated with the treatment group.

In addition, we conduct the *placebo test* by inserting several different "fake" policy implementation dates in the middle of the preintervention period. Specifically, we use the same pre-implementation sample (from October 1, 2014 to September 30, 2015), which only contains the time period before the actual policy intervention and test March 1, 2015, April 1, 2015, and May 1, 2015, respectively, as fictitious policy launch dates. Figure B1 shows the start and end date of our dataset, the date of the policy intervention, and the dates of the fictitious treatments adopted in the *placebo test*. The following regression specification is used:

$$\mathrm{DV}_{\mathrm{it}} = \zeta_1 \mathrm{Treatment\ Group}_i{}^{*}\mathrm{Policy}_t + \mu_i + W_t + \varepsilon_{\mathrm{it}} \quad \text{(B2)}$$

where the DV represents the six dependent variables in our main analysis. Results are presented in Tables B2, B3, and B4. Overall, the fictitious policy implementation dates did not exert a statistically significant effect on our dependent variables. Thus, the findings from the *placebo test* provide additional evidence to show that the treatment group and the control group had similar trajectories for the dependent variables before the policy implementation.

TABLE B3 Placebo test (April 1, 2015)

	Log (online sales)	Log (B&M sales)	Log (total sales)	Percentage of online sales from B&M unavailable products	Percentage of online orders using expedited shipping	Return rate
Treatment group × policy	0.0400	−0.0142	0.0101	0.0272	0.0550	−0.0552
	(0.0401)	(0.0145)	(0.0061)	(0.0201)	(0.3693)	(0.1330)
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Time effects	Yes	Yes	Yes	Yes	Yes	Yes
Propensity score matched sample	Yes	Yes	Yes	Yes	Yes	Yes
N	38,172	37,987	39,456	38,172	38,172	39,456
R square	0.202	0.103	0.163	0.115	0.076	0.022

Note: Clustered standard errors in parentheses. *$p < .05$; **$p < .01$; ***$p < .001$.

TABLE B4 Placebo test (May 1, 2015)

	Log (online sales)	Log (B&M sales)	Log (Total sales)	Percentage of online sales from B&M unavailable products	Percentage of online orders using expedited shipping	Return rate
Treatment group × policy	0.0093	−0.0320	−0.0123	0.0240	0.5630	0.0392
	(0.0143)	(0.0849)	(0.0064)	(0.0181)	(0.3502)	(0.1501)
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Time effects	Yes	Yes	Yes	Yes	Yes	Yes
Propensity score matched sample	Yes	Yes	Yes	Yes	Yes	Yes
N	38,172	37,987	39,456	38,172	38,172	39,456
R square	0.202	0.105	0.168	0.115	0.078	0.022

Note: Clustered standard errors in parentheses. *$p < .05$; **$p < .01$; ***$p < .001$.

APPENDIX C

C.1 | Robustness checks

In this section, we report the results of several robustness tests discussed in Section 6 to further validate our findings.

C.1.1. | Robustness checks on alternative sample constructions

We first investigate whether our results are sensitive to the way the sample is constructed. Tests for two additional sample construction methods are included in this section: (1) ZIP codes that generate a total of 169 orders or more during our 39-month (169 weeks) period of analysis (on average one order per week), and (2) ZIP codes that generate at least two orders each month. We reproduced the analysis for all of the dependent variables in our study with the two alternative samples. Results are presented in Tables C1 and C2, respectively. Our findings remain consistent when we use different sample construction criteria.

C.1.2. | Robustness checks on alternative data aggregation method

Second, we analyze alternative data aggregation methods to validate our results. Specifically, we use "ZIP Code - Week" as our unit of analysis instead of "ZIP Code—Month." We focus on ZIP codes that, on average, gener-

TABLE C1 Robustness analysis with alternative sample (4560 ZIP codes)

	ZIP codes that on average generate one order each week					
	Log (online sales)	Log (B&M sales)	Log (total sales)	Percentage of online sales from B&M unavailable products	Percentage of online orders using expedited shipping	Return rate
Treatment group × policy	0.116***	−0.040***	0.042***	5.674***	0.276+	0.184**
	(0.028)	(0.007)	(0.008)	(0.310)	(0.167)	(0.060)
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Time effects	Yes	Yes	Yes	Yes	Yes	Yes
Propensity score matched sample	Yes	Yes	Yes	Yes	Yes	Yes
N	159,560	158,149	174,484	159,560	159,560	174,484
R square	0.432	0.200	0.356	0.147	0.045	0.010

Note: Clustered standard errors in parentheses. $^*p < .05$; $^{**}p < .01$; $^{***}p < .001$; $+p < .1$.

TABLE C2 Robustness analysis with alternative sample (2598 ZIP codes)

	ZIP codes that generate at least two orders each month					
	Log (online sales)	Log (B&M sales)	Log (Total sales)	Percentage of online sales from B&M unavailable products	Percentage of online orders using expedited shipping	Return rate
Treatment group × policy	0.129***	−0.019+	0.037***	3.892***	0.207	0.137*
	(0.028)	(0.011)	(0.008)	(0.370)	(0.182)	(0.058)
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Time effects	Yes	Yes	Yes	Yes	Yes	Yes
Propensity score matched sample	Yes	Yes	Yes	Yes	Yes	Yes
N	99,214	98,414	101,322	99,214	99,214	101,322
R square	0.502	0.240	0.505	0.158	0.057	0.014

Note: Clustered standard errors in parentheses. $^*p < .05$; $^{**}p < .01$; $^{***}p < .001$; $+p < .1$.

ate one order each week. Results are presented in Table C3 and show that our findings are consistent using different data aggregation methods.

C.1.3. | Robustness checks on alternative regression estimation method

We also test the robustness of our results using a fixed-effects Poisson regression analysis. This count model is useful in handling non-negative yet skewed data and allows us to include a small proportion of channel-level observations with zero sales that are dropped when performing the logarithmic transformation. Results are presented in Table C4 and confirm that our findings are unbiased by the regression estimation method.

C.1.4. | Robustness checks on alternative trade area definitions

Fourth, we examine whether the results are sensitive to the trade area definition by adopting trade area definitions with a 10-mile, 20-mile, 30-mile, 40-mile, and 60-mile radius, respectively. Results based on each trade area definition are presented in Tables C5–C9, respectively. Figure C1 shows the treatment effect of each dependent variable across trade area definitions.

TABLE C3 Robustness analysis with alternative data aggregation method (4560 ZIP codes)

	ZIP codes that on average generate one order each week (weekly level analysis)					
	Log (online sales)	Log (B&M sales)	Log (total sales)	Percentage of online sales from B&M unavailable products	Percentage of online orders using expedited shipping	Return rate
Treatment group × policy	0.080***	−0.024***	0.040***	3.928***	0.243+	0.152**
	(0.022)	(0.006)	(0.005)	(0.728)	(0.134)	(0.052)
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Time effects	Yes	Yes	Yes	Yes	Yes	Yes
Propensity score matched sample	Yes	Yes	Yes	Yes	Yes	Yes
N	496,915	436,226	624,337	496,915	496,915	624,337
R square	0.540	0.230	0.510	0.162	0.178	0.010

Note: Clustered standard errors in parentheses. *$p < .05$; **$p < .01$; ***$p < .001$; +$p < .1$.

TABLE C4 Robustness analysis with alternative regression estimation method

	Poisson regression analysis					
	Log (online sales)	Log (B&M sales)	Log (total sales)	Percentage of online sales from B&M unavailable products	Percentage of online orders using expedited shipping	Return rate
Treatment group × policy	0.109***	−0.043***	0.024+	0.159***	0.040*	0.124***
	(0.020)	(0.010)	(0.013)	(0.011)	(0.020)	(0.024)
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Time effects	Yes	Yes	Yes	Yes	Yes	Yes
Propensity score matched sample	Yes	Yes	Yes	Yes	Yes	Yes
N	128,232	128,232	128,232	128,232	128,232	128,232
Log-pseudo likelihood	−18,709,105	−19,948,561	−22,635,171	−56,666	−1,156,190	−435,935

Note: Clustered standard errors in parentheses. *$p < .05$; **$p < .01$; ***$p < .001$; +$p < .1$.

TABLE C5 Robustness analysis with alternative trade area definition (10 miles)

	Log (online sales)	Log (B&M sales)	Log (Total sales)	Percentage of online sales from B&M unavailable products	Percentage of online orders using expedited shipping	Return rate
Treatment group × policy	0.175***	0.005	0.040***	2.185***	0.317	0.208*
	(0.016)	(0.017)	(0.012)	(0.530)	(0.278)	(0.092)
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Time effects	Yes	Yes	Yes	Yes	Yes	Yes
Propensity score matched sample	Yes	Yes	Yes	Yes	Yes	Yes
N	47,747	49,412	49,686	47,747	47,747	49,686
R square	0.419	0.142	0.348	0.153	0.052	0.010

Note: Clustered standard errors in parentheses. *$p < .05$; **$p < .01$; ***$p < .001$.

TABLE C6 Robustness analysis with alternative trade area definition (20 miles)

	Log (online sales)	Log (B&M sales)	Log (total sales)	Percentage of online sales from B&M unavailable products	Percentage of online orders using expedited shipping	Return rate
Treatment group × policy	0.140***	0.010	0.042***	2.786***	0.296	0.088+
	(0.033)	(0.011)	(0.008)	(0.393)	(0.183)	(0.049)
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Time effects	Yes	Yes	Yes	Yes	Yes	Yes
Propensity score matched sample	Yes	Yes	Yes	Yes	Yes	Yes
N	105,306	107,788	108,966	105,306	105,306	108,966
R square	0.441	0.328	0.383	0.155	0.060	0.015

Note: Clustered standard errors in parentheses. *$p < .05$; **$p < .01$; ***$p < .001$; $+p < .1$.

TABLE C7 Robustness analysis with alternative trade area definition (30 miles)

	Log (online sales)	Log (B&M sales)	Log (Total sales)	Percentage of online sales from B&M unavailable products	Percentage of online orders using expedited shipping	Return rate
Treatment group × policy	0.152***	−0.018+	0.055***	3.350***	0.371*	0.095+
	(0.029)	(0.010)	(0.007)	(0.371)	(0.169)	(0.056)
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Time effects	Yes	Yes	Yes	Yes	Yes	Yes
Propensity score matched sample	Yes	Yes	Yes	Yes	Yes	Yes
N	125,056	125,184	129,714	125,056	125,056	129,714
R square	0.441	0.457	0.410	0.156	0.059	0.013

Note: Clustered standard errors in parentheses. *$p < .05$; **$p < .01$; ***$p < .001$; $+p < .1$.

TABLE C8 Robustness analysis with alternative trade area definition (40 miles)

	Log (online sales)	Log (B&M sales)	Log (Total sales)	Percentage of online Sales from B&M unavailable products	Percentage of online orders using expedited shipping	Return rate
Treatment group × policy	0.146***	−0.019*	0.051***	3.896***	0.246+	0.142*
	(0.029)	(0.008)	(0.008)	(0.372)	(0.140)	(0.058)
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Time effects	Yes	Yes	Yes	Yes	Yes	Yes
Propensity score matched sample	Yes	Yes	Yes	Yes	Yes	Yes
N	126,220	125,214	130,416	126,220	126,220	130,416
R square	0.469	0.257	0.424	0.156	0.052	0.012

Note: Clustered standard errors in parentheses. *$p < .05$; **$p < .01$; ***$p < .001$; +$p < .1$.

TABLE C9 Robustness analysis with alternative trade area definition (60 miles)

	Log (online sales)	Log (B&M sales)	Log (total sales)	Percentage of online Sales from B&M unavailable products	Percentage of online orders using expedited shipping	Return rate
Treatment group × policy	0.115***	−0.032**	0.030***	4.838***	0.206*	0.125*
	(0.027)	(0.011)	(0.008)	(0.411)	(0.086)	0.060
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Time effects	Yes	Yes	Yes	Yes	Yes	Yes
Propensity score matched sample	Yes	Yes	Yes	Yes	Yes	Yes
N	124,846	122,214	129,246	124,846	124,846	129,246
R square	0.480	0.203	0.435	0.155	0.050	0.010

Note: Clustered standard errors in parentheses. *$p < .05$; **$p < .01$; ***$p < .001$.

C.1.5. | Robustness checks on possible impact of extreme transaction values

We next check whether our findings are biased by extreme transaction values. Specifically, we winsorize transaction values by the top and bottom 0.1% (20,892 transactions winsorized), 1% (214,183 transactions winsorized), and 5% (1,048,733 transactions winsorized), respectively. Results are presented in Tables C10–C12 and remain consistent with our main findings.

C.1.6. | Robustness checks on the inclusion of demographic variables

Sixth, we test whether our findings are robust when we directly include demographic variables used in sample matching as control variables in our regression analysis. We add population, median household income, unemployment rate, and total retail sales as control variables to our regression analysis. Results are presented in Table C13 and indicate that our findings are consistent with the inclusion of additional demographic control variables.

C.1.7. | Robustness checks on alternative matching criteria

And lastly, we conduct robust analysis to investigate whether our findings are sensitive to the caliper value adopted when performing the propensity score matching. We test for two additional caliper values, 0.5 and 0.05, respectively, to verify whether our findings are affected by this value. Specifically, when using a caliper of 0.5, 1946 out of 2825 ZIP codes from the treatment group have a suitable match in the control group. Alternatively, when using a caliper of 0.05, 1569 out of 2825 ZIP codes from the treatment group have a suitable match. Table C14 reports the demographic variable comparisons between the treatment and control groups under the two caliper values. Tables C15 and C16 present our main analysis with the new caliper values and demonstrate consistency.

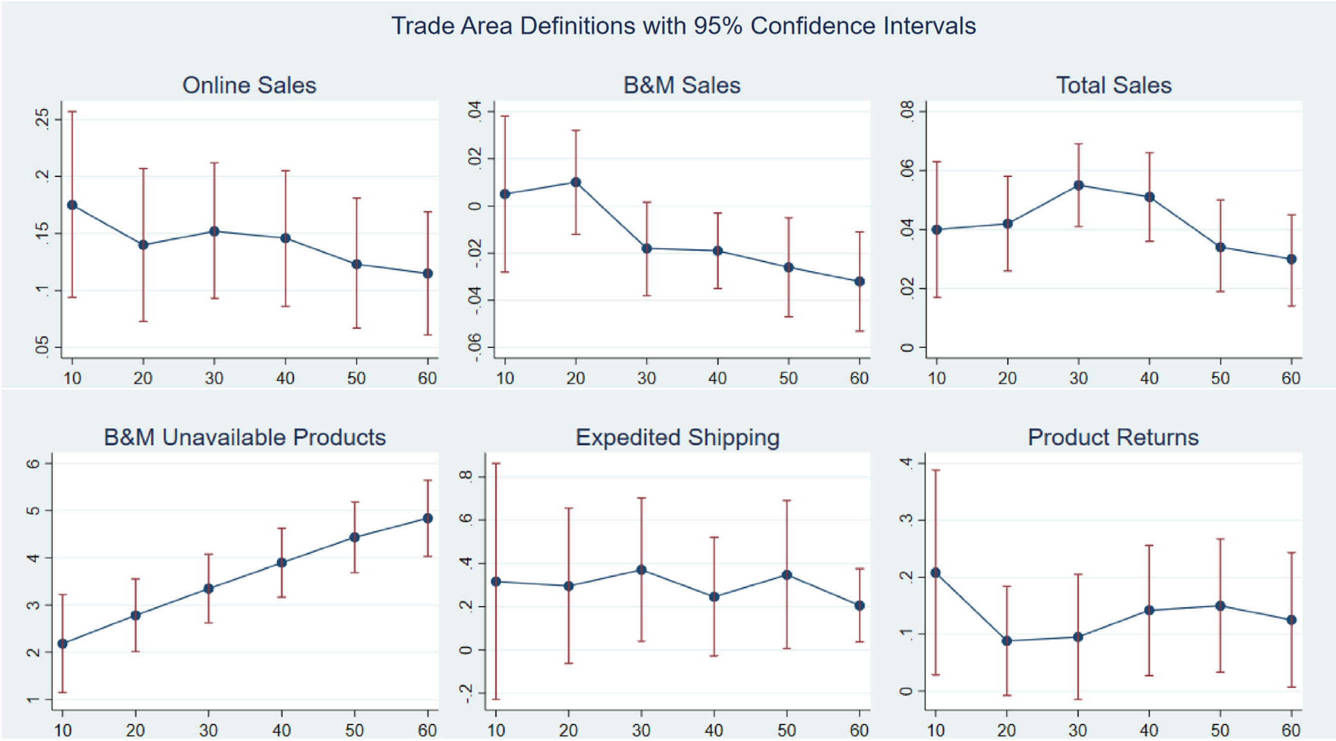

FIGURE C1 Treatment effect based on trade area definitions. Values on the *X*-axis represent different trade area definitions. Values on the *Y*-axis represent treatment effect with 95% confidence intervals.

TABLE C10 Robustness analysis with extreme transaction values excluded (basket value winsorized at 0.1%)

	Basket value winsorized at top and bottom 0.1%					
	Log (online sales)	**Log (B&M sales)**	**Log (total sales)**	**Percentage of online Sales from B&M unavailable products**	**Percentage of online orders using expedited shipping**	**Return rate**
Treatment group × policy	0.128***	−0.028**	0.037***	4.432***	0.385*	0.150**
	(0.029)	(0.011)	(0.006)	(0.380)	(0.176)	(0.058)
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Time effects	Yes	Yes	Yes	Yes	Yes	Yes
Propensity score matched sample	Yes	Yes	Yes	Yes	Yes	Yes
N	123,618	121,838	128,231	123,618	123,618	128,231
R square	0.482	0.217	0.436	0.158	0.051	0.010

Note: Clustered standard errors in parentheses. *$p < .05$; **$p < .01$; ***$p < .001$.

TABLE C11 Robustness analysis with extreme transaction values excluded (basket value Winsorized at 1%)

	Basket value Winsorized at top and bottom 1%					
	Log (online sales)	Log (B&M sales)	Log (total sales)	Percentage of online Sales from B&M unavailable products	Percentage of online orders using expedited shipping	Return rate
Treatment group × policy	0.130***	−0.033**	0.036***	4.435***	0.384*	0.151**
	(0.028)	(0.010)	(0.006)	(0.381)	(0.175)	(0.058)
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Time effects	Yes	Yes	Yes	Yes	Yes	Yes
Propensity score matched sample	Yes	Yes	Yes	Yes	Yes	Yes
N	123,499	121,698	128,215	123,499	123,499	128,215
R square	0.482	0.218	0.438	0.156	0.051	0.015

Note: Clustered standard errors in parentheses. $^{*}p < .05$; $^{**}p < .01$; $^{***}p < .001$.

TABLE C12 Robustness analysis with extreme transaction values excluded (basket value winsorized at 5%)

	Basket value winsorized at top and bottom 5%					
	Log (online sales)	Log (B&M sales)	Log (total sales)	Percentage of online sales from B&M unavailable products	Percentage of online orders using expedited shipping	Return rate
Treatment group × policy	0.134***	−0.045***	0.038***	4.440***	0.407*	0.149**
	(0.027)	(0.010)	(0.006)	(0.383)	(0.175)	(0.058)
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Time effects	Yes	Yes	Yes	Yes	Yes	Yes
Propensity score matched sample	Yes	Yes	Yes	Yes	Yes	Yes
N	122,950	120,916	128,100	122,950	122,950	128,100
R square	0.486	0.216	0.439	0.150	0.052	0.015

Note: Clustered standard errors in parentheses. $^{*}p < .05$; $^{**}p < .01$; $^{***}p < .001$.

TABLE C13 Robustness analysis with demographic variables

	Log (online sales)	Log (B&M sales)	Log (Total sales)	Percentage of online sales from B&M unavailable products	Percentage of online orders using expedited shipping	Return rate
Treatment group × policy	0.111***	−0.028**	0.023**	4.154***	0.429*	0.152*
	(0.027)	(0.011)	(0.008)	(0.451)	(0.176)	(0.060)
Population (000)	0.006	−0.017**	0.004	0.006	0.000	−0.008
	(0.005)	(0.006)	(0.004)	(0.008)	(0.081)	(0.028)
Median household income (000)	−0.010	−0.010***	−0.009***	0.043	0.075	0.013
	(0.005)	(0.002)	(0.002)	(0.053)	(0.041)	(0.015)
Total retail sales (millions of $)	−0.000	0.000*	−0.000	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Unemployment rate (%)	−0.011*	0.007**	−0.011***	0.011	0.063	−0.020
	(0.005)	(0.003)	(0.002)	(0.021)	(0.038)	(0.015)
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Time effects	Yes	Yes	Yes	Yes	Yes	Yes
Propensity score matched sample	Yes	Yes	Yes	Yes	Yes	Yes
N	123,623	121,945	128,232	123,623	123,623	128,232
R square	0.428	0.216	0.391	0.202	0.055	0.010

Note: Clustered standard errors in parentheses. *$p < .05$; **$p < .01$; ***$p < .001$.

TABLE C14 Demographic statistics after propensity score matching

	PSM with caliper of 0.5			PSM with caliper of 0.05		
	Treatment	Control	*Z*-score	Treatment	Control	*Z*-score
Population	27,547	26,221	−2.8	26,606	26,778	0.3
Median household income	77,121	73,708	−5.3	74,091	75,616	2.1
Total retail sales (millions of $)	505	488	−1.0	490	494	0.3
Unemployment rate (%)	4.9	4.9	0.6	4.9	4.9	−1.0
Number of ZIP codes	1946	1946		1569	1569	

TABLE C15 Robustness analysis with alternative matching criteria

	Propensity score matching caliper 0.5 (3892 ZIP codes)					
	Log (online sales)	Log (B&M sales)	Log (total sales)	Percentage of online sales from B&M unavailable products	Percentage of online orders using expedited shipping	Return rate
Treatment group × policy	0.121***	−0.032**	0.036***	4.443***	0.376*	0.128*
	(0.028)	(0.010)	(0.007)	(0.361)	(0.159)	(0.054)
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Time effects	Yes	Yes	Yes	Yes	Yes	Yes
Propensity score matched sample	Yes	Yes	Yes	Yes	Yes	Yes
N	146,741	144,175	151,788	146,741	146,741	151,788
R square	0.488	0.216	0.433	0.158	0.052	0.011

Note: Clustered standard errors in parentheses. *$p < .05$; **$p < .01$; ***$p < .001$.

TABLE C16 Robustness analysis with alternative matching criteria

	Propensity score matching caliper 0.05 (3138 ZIP codes)					
	Log (online sales)	**Log (B&M sales)**	**Log (total sales)**	**Percentage of online sales from B&M unavailable products**	**Percentage of online orders using expedited shipping**	**Return rate**
Treatment group × policy	0.125***	−0.024*	0.037***	4.452***	0.388*	0.139*
	(0.029)	(0.011)	(0.008)	(0.382)	(0.179)	(0.061)
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Time effects	Yes	Yes	Yes	Yes	Yes	Yes
Propensity score matched sample	Yes	Yes	Yes	Yes	Yes	Yes
N	117,883	113,238	122,382	117,883	117,883	122,382
R square	0.468	0.215	0.426	0.152	0.051	0.010

Note: Clustered standard errors in parentheses. $^{*}p < .05$; $^{**}p < .01$; $^{***}p < .001$.